



CHEMISTRY Department
SIMHADRI

Note No :1



Ref: SMTTP/Chemistry/2023-24/ofc note/912558

Date:

15/02/2024(12:45:55)

Subject: LMI on Examination and Maintenance of Lubricating oils of Turbine and Boiler fill Pumps:

Sub: LMI on Examination and Maintenance of Lubricating oils of Turbine and Boiler fill Pumps:

LMI on Examination and Maintenance of Lubricating oils of Turbine and Boiler fill Pumps

LMI/OGN/OPS/CHEM/009 has been made as per the new TC Document

COS-ISO-00-OGN/OPS/CHEM/021 Rev. No.: 0 approved on 31 Jan. 2024 and following has been incorporated.

S. no	New	Changes Done as per
1.	Monitoring Frequency of BFP Oils have been incorporated. Section 5.0 Page no.6	COS-ISO-00-OGN/OPS/CHEM/021 Rev. No.: 0 Nov. 2023
2	Turbine Lub Oil Specifications Annexure -1 Page no.9 & Page no.10	COS-ISO-00-OGN/OPS/CHEM/021 Rev. No.: 0 Nov. 2023

3	Lub Oil Flushing Clearance parameters (post AOH)- separate section incorporated. Section 6.0 Page no.7	COS-ISO-00- OGN/OPS/CHEM/021 Rev. No.: 0 Nov. 2023
4	Non-compatibility of Turbo Generator Lubricating/ Oils Supplied by Different Oil Companies. Section 7.0 Page no.7	COS-ISO-00- OGN/OPS/CHEM/021 Rev. No.: 0 Nov. 2023
5	Records to be maintained for Analysis and Top up has been incorporated. Section 8.0 Page no.8	COS-ISO-00- OGN/OPS/CHEM/021 Rev. No.: 0 Nov. 2023
5	Compliance Checklist is incorporated. Annexure-III, Page no.12	COS-ISO-00- OGN/OPS/CHEM/021 Rev. No.: 0 Nov. 2023
6	Comparison table between NAS and ISO has been incorporated. Annexure – IV Page no.13	COS-ISO-00- OGN/OPS/CHEM/021 Rev. No.: 0 Nov. 2023

Submitted for Kind Approval and implementation.

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Note No:5

The Title of the LMI ("The Examination and Maintenance of Lubricating Oil of Turbine and Boiler fill Pumps"), may please be corrected as "The Examination and Maintenance of Lubricating Oil of Turbine and Boiler feed Pumps"

Please review.

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Note No:6

Reviewed and No changes proposed in the document.

Title of the LMI correction may be done.

Forwarded for further approval.

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Pl review the LMI

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Note No:9

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SIMHADRI SUPER THERMAL POWER PROJECT
OPERATION DIVISION FORMAL DOCUMENTATION SYSTEM
LOCATION MANAGEMENT INSTRUCTION- LMI/OGN/OPS/CHEM/009

ISSUE No: 02, Date: 01 February 2024

REVISION -04

The Examination and Maintenance of Lubricating Oil of Turbine and Boiler fill Pumps



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The Examination and Maintenance of Lubricating Oil of Turbine and Boiler fill Pumps
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1.0 INTRODUCTION

The turbine lubricating oil must withstand continuous thermal cycling and a variable degree of contamination by solid particles, sludge, water, hydrogen and air. It is required to release readily such contaminants and retain its lubricating and other properties over very long periods as total replacement of oil is prohibitively costly.

In order to meet the above objectives, periodic examination and maintenance of turbine lubricating oil is necessary.

2.0 SUPERSEDED DOCUMENTS:

LMI/OGN/OPS/CHEM/009 REVISION-03 DATED MAY-2022

3.0 SCOPE

In order to provide assurance that the oil is being maintained in a healthy condition, it is necessary to monitor selected properties of that oil.

This LMI defines limits for various parameters and frequency of testing of oils so that degradation in service does not result in undue threat to the machine.

4.0 SPECIFIC INSTRUCTIONS TO BE FOLLOWED

New Oil		
Sl no	Activity	Responsibility
4.1	All supplies should either be delivered with a BIS marking along with the procurement specification. If BIS certificate /marking is not available, samples may be taken and examined for appearance and viscosity before topping up. All samples should be within the permitted viscosity range and should appear clear and bright and visually free from solid and liquid contaminants.	HOD(CHEM) & HOD(TMD)
4.2	In the event of a disagreement over the appearance of the oil samples the presence of contaminants shall be determined. Maximum concentrations of 0.0007% weight by weight (7mg/l) insoluble inorganic material and 10mg/l water are recommended for new oil. Sampling should be done as per Operation Information Note titled "Manual on Sampling and Testing Procedure for Oil: COS-ISO-00-OIN/OPS/CHEM/015".	HOD(CHEM)
OIL IN SERVICE		
4.3	Turbine oils will gradually deteriorate in service due to oxidation and the presence of water, corrosion and wear products. Purifying equipment (i.e. Centrifuge etc.) should be pressed into service to maintain the quality of the oil supplied to the bearings.	HOD(OPN)

4.4	Sampling and monitoring should be carried out on a regular basis to ascertain the rate of deterioration and to enable necessary corrective measures to be taken.	HOD(CHEM)
4.5	During normal Unit operation: Representative samples of the oil being delivered to the machine should be obtained from any convenient point preferably from turbulent flow areas such as turbine bearing return lines, when the oil is in circulation, and tested as specified in Table-II.	HOD(CHEM)
	When the unit is coming after overhauling, during the turbine lub oil flushing activity, oil samples should be taken from Oil tank inlet and outlet line for monitoring the parameters and clearance is to be accorded as per the limits mentioned in Table-III (Sec 10.0). It may be desirable to obtain samples at different positions in order to trace sources of contamination as per the requirement. For detail sampling procedure, the Operation Information Note titled "Manual on Sampling and Testing Procedure for Oil: COS-ISO-00-OIN/OPS/CHEM/015", should be referred to.	HOD(CHEM)
TREATMENT AND MAINTENANCE		
4.6	Water, dirt, hydrogen and air should be removed from the oil as efficiently and as quickly as the system design allows using the installed purification equipment. It is desirable to use condensate or de-mineralized water for sealing centrifuges and operation of these at a temperature of at least 65 deg. C is normally required to break oil / water emulsions. Wet and / or dirty oil returned to the turbine will eventually manifest itself in corrosion, wear malfunctioning of the equipment and biological growth.	HOD(TMD)
4.7	Water and water / oil emulsion should be drained from the main oil tank drain at intervals to ensure that a low level of water is maintained in the tank. Accumulated water or water oil emulsion shall be drained from lowest drain of the main oil tank at available opportunities like unit shut down.	HOD(OPN)
4.8	Centrifuge should be in service with heater temperature set at 65- 80 deg C to break oil emulsion. Heater temperature should not exceed 80 deg C to avoid oxidation of oil. It is desirable to use condensate or de-mineralized water for sealing centrifuges. Vapor extraction fans shall be in service continuously to remove any moisture vapors.	HOD(OPN)

4.9	Centrifuge bowl should be cleaned to keep it in clean Condition during 4 weekly preventive maintenance.	HOD(TMD)
CONDITION MONITORING OF OIL IN SERVICE		
4.10	Periodic monitoring of the condition of the lubricating oil of Turbine and BFP in each operational unit should be carried out in accordance with Table-I & Table-II. If parameters are outside the recommended range, then the reasons for this should be sought and remedial action taken, as deemed fit.	HOD(CHEM) & HOD(TMD)
4.11	The consequences of continued operation with oil outside the specified condition is detrimental to the health of the equipment and it may be necessary to consult the oil supplier to discuss the need for an oil change or refurbishment of the oil by the use of an additive package.	HOD(TMD)
4.12	Condition monitoring of turbine oil for various parameter should be carried out in specified frequency as detailed in Annexure-I. Data should be uploaded in intranet, Chemistry Website. Deviations in parameter should be reported in DPM meeting	HOD(CHEM)
4.13	Lub oil samples should be sent separately to CC-NETRA (six-monthly) for detailed analysis and recommendation.	HOD (CHEM)

But for taking any decision, OEM/oil supplier's recommendations must be considered

5.0 The frequency of testing/ monitoring of various parameters on routine basis shall be as follows:

Table-IA: Turbine Lubricating Oil			
Sl.No.	Parameters	UOM	Testing Frequency
1	Moisture	ppm	Once/Week
2	Kinematic Viscosity	cst	Once/Week
3	Total Acid number	mg KOH/g of oil	Fortnightly
4	Mechanical Impurity	ppm	Once/Week
5	Particle count and size as per NAS 1638 and/or ISO 4406	----	Once/month

BFP Oils

Table-IB: MDBFP Lub Oil

Sl.No.	Parameters	UOM	Testing Frequency
1	Moisture	ppm	Once/Week
2	Kinematic Viscosity	cst	Once/Week
3	Total Acid number	mg KOH/g of oil	Fortnightly
4	Mechanical Impurity	ppm	Once/Week
5	Particle count and size as per NAS 1638 and/or ISO 4406	----	Once/month

Table-IC: TDBFP Lub Oil

Sl.No.	Parameters	UOM	Testing Frequency
1	Moisture	ppm	Once/Week
2	Kinematic Viscosity	cst	Once/Week
3	Total Acid number	mg KOH/g of oil	Fortnightly
4	Mechanical Impurity	ppm	Once/Week
5	Particle count and size as per NAS 1638 and/or ISO 4406	----	Once/month

Recommendations:

	Activity	Responsibility
5.1	The recommended frequency of the above-mentioned tests may be increased if there is persistent aberration in the testing parameters to ascertain the cause of deviation from normal parameter range. Oil coolers surface and other associated equipments may also be inspected to bring and maintain the parameters within the recommended limits.	AGM(TMD) & AGM(CHEM)
5.2	The moisture content of turbine lubricating oil varies widely with plant design and operational practices, although, it is desirable to keep the moisture content as low as practicable and always below 100 mg/l. If this is not achievable with some plant, reasons for such excursions should be sought and remedial action taken. More frequent monitoring may be necessary under these conditions to bring down the value to less than 100 mg/l.	AGM(TMD) & AGM(CHEM)

6.0TURBINE LUB OIL FLUSHING PARAMETERS MONITORING (POST ANNUALOVERHAULING)

Table-III				
Sl.No.	Parameters	UOM	Limits	Remarks
1	Moisture	ppm	100	In-line filtration is in clean condition and no choking of filter has happened during 04 hours of oil flushing activity.
2	Mechanical Impurity	ppm	100	

7.0Non-Compatibility of Turbo Generator Lubricating Oils Supplied by Different Oil Companies:

Control Procedures		
	Activity	Responsibility
7.1	stations may consider topping up systems with oil of the same specification but from a different manufacturer. Only oil of recommended grade by equipment manufacturer must be used. However, if, for any reason, the original grade of oil is not available for topping-up, then the full charge of oil must be replaced with a direct alternative of the same grade.	AGM (TMD)
7.2	a) manufacturer or, b) any Standard Research Laboratory such as 1)CSIR Lab 2)National Test Houses. 3)M/s IOCL, Faridabad etc should be approached for declaring compatibility of new oil with the existing oil, available as main bulk, filled inside the system. If it is found compatible, then only replacement can be planned.	AGM(CHEM)
7.3	The lubricating oil Control oil/Seal oil must only be topped-up with oil of the same type, grade and manufacturer as the original oil. If oil of the original type is not available for toping-up then the full charge must be replaced, taking care that all the original oil is removed.	AGM(OPRN)

Records to Maintained:

Sl No.	Record Description	Maintained by	Frequency	Remarks
A. For routine lub oil analysis:				
1	Test results of routine analysis of lub oil samples	Chemistry	As per table-IA, IB, IC	
2	External test reports of CC-NETRA	Chemistry (external test report copies to be available with site TMD)	Six-Monthly or earlier if required	
3	External test reports of OEM/Supplier	Turbine Maintenance (external test report copies to be available with site Chemistry)	Yearly or earlier if required	
B. For Oil replacement/Top-up:				
1	Records must be kept for all inspections and should contain the following data: Details of oil top-up Details of oil replacement & reasons Quality checks of oil for replacement	Operation/ TMD as applicable - do- Chemistry	As per requirement	

Annexure -I

Turbine Lubricating Oil Specifications- Sub Critical Units

Table-IIA: Turbine Lubricating Oil Specification for ≤500 MW Units (as per OEM specification and IS 1012:2002, Reaffirmed-2018- Turbine Lubricating Oil Specifications)

Sl no	Characteristics/ Parameter	Limits (VG 32)	Limits (VG 46)	Testing Frequency	Test method
1.	Moisture Content mg/l	<100 (Note ii)	<100 (Note ii)	Weekly (Note ii)	ASTM D6304
2.	Kinematic Viscosity at 40 deg C, centistokes (cst or mm ² /s)	28.8 – 35.2	41.4 – 50.6	Weekly	IS:1448, P:25 ASTM D445
3.	Viscosity index, min	100	98	Six Monthly#	IS:1448, P:56 ASTM D2270
4.	Density at 15 deg C (kg/M ³)	Max 900	Max 900	Six Monthly#	ASTM D1298
5.	Flash point (Cleveland open cup.) 0 C min	190	200	Six Monthly#	IS:1448, P:69 ASTM D92
6.	Copper strip corrosion at 100 deg. C for three hours	Not worse than No. 1	Not worse than No. 1	Six Monthly#	IS:1448, P:15 ASTM D130
7	Rust preventive characteristics, 24 hr test	Shall pass the test	Shall pass the test	Six Monthly#	IS:1448, P:96 ASTM D665
8.	Emulsion characteristics, Max	40-40-0 (20)	40-40-0 (20)	Six Monthly#	IS:1448, P:91 ASTM D1401
9.	Total Acid Number (TAN), mg KOH/g oil	≤0.2	≤0.2	Fortnightly	IS:1448, P:188 ASTM D974
10.	Foaming at 25 deg C Tendency (cm ³)	Max 400	Max 400	Six Monthly#	IS:1448, P:67 ASTM D892

Table-IIA: Turbine Lubricating Oil Specification for ≤500 MW Units (as per OEM specification and IS 1012:2002, Reaffirmed-2018- Turbine Lubricating Oil Specifications)

Sl no	Characteristics/ Parameter	Limits (VG 32)	Limits (VG 46)	Testing Frequency	Test method
	Stability (s)	Max 450	Max 450		
11.	Air release value, at 50 deg. C. Max in minutes.	5	6	Six Monthly#	IS:1448, P:102 ASTM D3427
12.	Mechanical impurities % (solid particles by weight)	<0.01	<0.01	Weekly	ASTM D4055
13.	Particle Distribution	≤19/17/14 (ISO 4406) NAS Class 8 (max as per NAS 1638) (Note iii)	≤19/17/14 (ISO 4406) NAS Class 8 (max as per NAS 1638) (Note iii)	Monthly	ISO 4406 NAS 1638
14.	Accelerated ageing test (RPVOT), Minutes (Min)	200 (or OEM recommended)	200 (or OEM recommend ed)	Six Monthly#	ASTM D 2272
15.	Colour (max)	≤2.0	≤2.0	Six Monthly#	IS:1448, P:12 ASTM D1500

For the test methods as mentioned in the Table, latest version of the relevant Standards (IS/ASTM) should be considered. The limits for the above are based on the OEM specifications/ IS 1012 (latest version). In case of any mismatch in lub oil parameter limits between OEM recommendation and IS 1012 standard, OEM recommended limit must be taken into consideration.

Annexure- II

Interpretation Chart of Testing Parameters

Sl. No	Testing Parameter	Limits	Interpretation	Action
1	Moisture	As per Table -IIA & IIB	1. More susceptible to oxidation and degradation 2. Enhance corrosion 3. Deplete additives 4. Risk of bacterial/fungal growth 5. Increase moisture in Hydrogen used for generator coolant	Investigate/ check functioning of centrifuge equipment.
2	Viscosity	-do-	1. Oil contaminated 2. Oil severely degraded.	Consider for oil change.
3	Acidity	-do-	Oil degraded. Oil contaminated High acidity will increase corrosion potential. Catalyze further degradation reactions.	Increase frequency of testing & investigate cause. Consider oil change.
4	Flash point	-do-	1. Oil contaminated 2. Oil severely degraded	Increase frequency of testing & investigate cause. Consider oil change
5	Rust Preventive characteristics	-do-	The system is very severe on oil, i.e. it is very wet and/or dirty. The system is incorrectly maintained, i.e. water drainage neglected and/ or centrifuge not operating satisfactorily. Normal inhibitor depletion in wet system.	Investigate cause immediately and rectify. Recheck oil and consult oil company regarding re-inhibition if test result is unchanged
6	Emulsion Characteristics	-do-	Oil degraded. Oil contaminated Oil coolers leakage	Investigate functioning of oil purification equipment / centrifuge Check oxidation stability. Check rusting characteristics

7	Mechanical impurities	-do-	1. Purification equipment not functioning properly. 2. Severe oil degradation is occurring. 3. Oil contamination is occurring. 4. Additive Precipitation is occurring. 5. Indication of presence of Corrosion and/or debris.	Centrifuge bowl cleaning
8.	Rotating pressure Vessel Oxidation test	-do-	1. Determines oxidation stability of oils. 2. Measures actual resistance to oxidation of oils. 3. Performance of added anti-oxidants in oil.	Estimate of fluid health and provide estimate of remaining life.

Annexure- III

Checklist for OGN-OPS-CHEM-009, Rev No-05 on "The Examination and Maintenance of Lubricating Oils of Turbine & Boiler Fill Pumps"

SI No	Item description	Reference clause/section in OGN/OPS/CHEM/009	Responsibility	Compliance (Yes/No)
1.	Compliance check for new oil specifications	Section 4.0	TMD	Yes/No
2.	Lub oil sample collection	Section 6.0	Chemistry in presence of Operation/TMD	Yes/No
3.	Treatment and maintenance of oil	Section 7.0	TMD	Yes/No
4.	Condition monitoring of oil in service	Section 8.0, 9.0	Chemistry (in-house) & CC-NETRA (external) TMD (for external testing from OEM/ Supplier)	Yes/No
5.	Oil flushing parameters	Section 10.0	Chemistry	Yes/No

	monitoring & clearance			
6.	Oil Top-up and replacement as per compatibility	Section 12.0	Operation (for top-up) Chemistry (for testing)	Yes/No
7	Compatibility checks of new oil	Section 12.0	Operation	Yes/No

Annexure -IV

Comparative table between NAS-1638 and ISO:4406 (based on maximum contamination limits and particles per 100 ml):

Approximate ISO 4406 Equivalent	NAS 1638 Code	5-15 μm	15-25 μm	25-50 μm	50-100 μm	over 100 μm
-	0	125	22	4	1	0
-	0	250	44	8	2	0
12/10/7	1	500	89	16	3	1
13/11/8	2	1000	178	32	6	1
14/12/9	3	2000	356	63	11	2
15/13/10	4	4000	712	126	22	4
16/14/11	5	8000	1425	253	45	8
17/15/12	6	16000	2850	506	90	16
18/16/13	7	32000	5700	1012	190	32
19/17/14	8	64000	11400	2025	360	64
20/18/15	9	128000	22800	4050	720	128
21/19/16	10	256000	45600	8100	1440	256
22/20/17	11	512000	91200	16200	2880	512
23/21/18	12	1024000	182400	32400	5760	1020

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9.0 Review:

Head of Chemistry is responsible for reviewing the document on a 2-yearly basis and as required.